

Comparative Analysis of Neuroinflammation-Associated Gene Expression in TCGA Lower-Grade Glioma and Glioblastoma

Abstract

Background: Neuroinflammation-associated genes may exhibit different expression patterns across glioma tumour groups. This exploratory study compared the expression values of ten selected genes between The Cancer Genome Atlas (TCGA) lower-grade glioma (LGG) and glioblastoma (GBM) primary tumour samples.

Methods: Gene-expression data were obtained from the UCSC Xena TCGA lower-grade glioma and glioblastoma dataset ([TCGA.GBMLGG.sampleMap/HiSeqV2](#)) and matched with clinical annotations. After restricting the analysis to primary tumour samples, 516 LGG and 154 GBM samples were included. Expression distributions were compared using two-sided Mann–Whitney U tests, with p-values adjusted across the ten gene-level comparisons using the Benjamini–Hochberg procedure. Cliff’s delta was calculated with positive values indicating a tendency toward higher expression values in GBM.

Results: Seven genes showed statistically significant differences in expression distributions after multiple-testing correction. CXCL10, CASP1, PYCARD, IL18, AIF1 and IL1B had higher median expression values in GBM than in LGG, whereas TLR4 had a lower median. TNF, NLRP3 and CSF1R did not meet the adjusted significance threshold. CXCL10 showed the largest observed median difference (GBM minus LGG = 4.3913; adjusted $p = 3.78 \times 10^{-50}$; Cliff’s $\delta = 0.798$).

Conclusions: This exploratory analysis identified differences in dataset-provided expression values for several selected genes between LGG and GBM primary tumour samples. These group-level associations do not establish causation, identify the cellular source of expression signals, or demonstrate longitudinal tumour progression. Independent validation and analyses accounting for relevant clinical and molecular covariates are needed.

Keywords

Glioma; glioblastoma; lower-grade glioma; neuroinflammation; gene expression; TCGA; UCSC Xena

1. Introduction

Gliomas are a diverse group of tumours of the central nervous system. Lower-grade glioma (LGG) and glioblastoma (GBM) represent clinically and biologically distinct tumour groups. Large-scale molecular studies, including The Cancer Genome Atlas (TCGA), have helped characterize the genomic and molecular diversity of gliomas (Cancer Genome Atlas Research Network, 2015).

Inflammatory signalling and immune-associated processes are relevant areas of investigation in tumour biology. Genes involved in inflammasome-related signalling, cytokine pathways, innate immune responses and myeloid-cell-associated biology may show different expression patterns across tumour samples. However, bulk-tumour expression measurements reflect a mixture of cell types and cannot, by themselves, establish which cells produced a given transcript or whether expression differences have a causal role in tumour biology.

The present study performs an exploratory comparison of ten selected neuroinflammation-associated genes, NLRP3, PYCARD, CASP1, IL1B, IL18, TLR4, TNF, CXCL10, AIF1 and CSF1R, between TCGA LGG and GBM primary tumour samples. The research question was: How do expression patterns of selected neuroinflammation-associated genes differ between TCGA lower-grade glioma and glioblastoma?

2. Materials and Methods

2.1 Data source and sample selection

Gene-expression data were obtained from the UCSC Xena platform using the TCGA lower-grade glioma and glioblastoma (GBMLGG) dataset, identified as [TCGA.GBMLGG.sampleMap/HiSeqV2](#) (UCSC Xena Accessed on Oct 4th, 2026). The dataset metadata describes the data type as RNA-seq gene expression and the expression unit as $\log_2(\text{norm_count} + 1)$. Clinical annotations were used to identify tumour type and sample type. The analysis was restricted to primary tumour samples classified as LGG or GBM. After filtering and matching samples between the expression and clinical datasets, 516 LGG samples and 154 GBM samples were included (670 samples total). The dataset contains 702 samples overall, which is distinct from the final analysis set of 670 primary tumour samples.

The expression values were used as provided in the downloaded dataset, whose metadata report the unit as $\log_2(\text{norm_count} + 1)$. The upstream normalization procedure was not independently validated as part of this analysis.

2.2 Candidate genes

Ten neuroinflammation-associated candidate genes were included in the exploratory analysis: NLRP3, PYCARD, CASP1, IL1B, IL18, TLR4, TNF, CXCL10, AIF1 and CSF1R. The panel covered genes associated with inflammasome-related signalling, cytokine pathways, innate immune responses and myeloid-cell-associated biology. The analysis was intended as a focused comparison of selected candidates rather than a comprehensive assessment of all neuroinflammation-associated genes.

2.3 Statistical analysis

For each candidate gene, the median expression value was calculated separately for LGG and GBM, and the median difference was defined as the GBM median minus the LGG median. Expression distributions were compared using two-sided Mann-Whitney U tests with the asymptotic method. The resulting p-values were adjusted across the ten gene-level comparisons using the Benjamini-Hochberg procedure. An adjusted p-value below 0.05 was considered statistically significant.

Cliff's delta was calculated with the direction GBM minus LGG; positive values indicate a tendency toward higher expression values in GBM, whereas negative values indicate a tendency toward higher values in LGG. The Mann-Whitney U test assesses differences between distributions and is not, without additional assumptions, a test of medians alone. Statistical analyses were performed in Python using SciPy for the Mann-Whitney U tests and statsmodels for multiple-testing correction. The software and package versions should be reported for reproducibility.

3. Results

3.1 Sample composition

Following sample-type filtering and matching to clinical annotations, the analysis included 516 LGG primary tumour samples and 154 GBM primary tumour samples, for a total of 670 samples.

3.2 Differences in candidate-gene expression values

Seven of the ten candidate genes showed statistically significant differences in expression distributions between LGG and GBM after Benjamini-Hochberg correction (adjusted $p < 0.05$). Six genes had higher median expression values in GBM than in LGG: CXCL10, CASP1, PYCARD, IL18, AIF1 and IL1B. CXCL10 showed the largest observed difference, with a median of 3.24335 in LGG and 7.63465 in GBM (median difference = 4.39130; Cliff's delta = 0.798148; adjusted $p = 3.784614 \times 10^{-50}$).

CASP1, PYCARD, IL18, AIF1 and IL1B also had positive Cliff's delta values, indicating a tendency toward higher expression values in GBM. TLR4 showed a lower median in GBM than in LGG (9.57080 versus 9.85890; Cliff's delta = -0.187620 ; adjusted $p = 5.799544 \times 10^{-4}$).

The remaining three genes, TNF, NLRP3 and CSF1R, did not show statistically significant differences after multiple-testing correction, with adjusted p -values of 0.209290, 0.262637 and 0.488100, respectively. The complete statistical results are presented in Table 1, and the gene-expression distributions are shown in Figure 1.

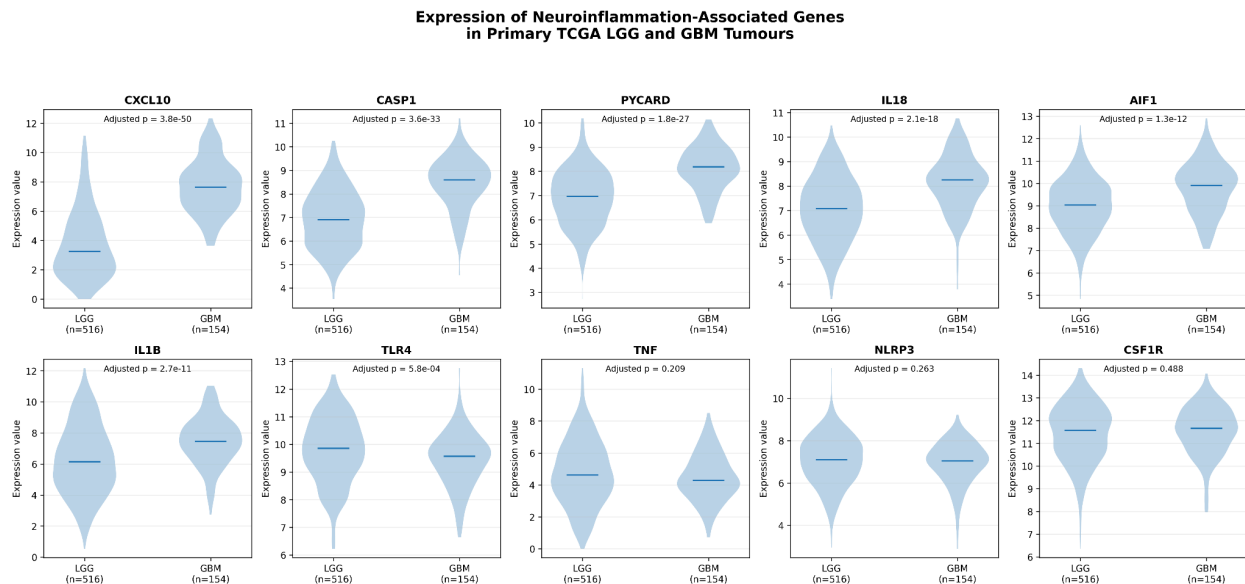


Figure 1. Expression of neuroinflammation-associated genes in primary TCGA lower-grade glioma (LGG) and glioblastoma (GBM) tumours. Violin plots compare expression values for ten candidate genes between LGG ($n = 516$) and GBM ($n = 154$) primary tumour samples. Horizontal lines indicate group medians. Adjusted p -values are from two-sided Mann-Whitney U tests with

Benjamini-Hochberg correction across ten gene-level comparisons. The UCSC Xena dataset metadata reports expression values in $\log_2(\text{norm_count} + 1)$ units.

Table 1. Comparison of neuroinflammation-associated gene expression between LGG and GBM.

Gene	LGG median	GBM median	Median difference*	Cliff's δ	Raw p-value	Adjusted p-value
CXCL10	3.243	7.635	4.391	0.798	3.78×10^{-51}	3.78×10^{-50}
CASP1	6.903	8.596	1.693	0.644	7.18×10^{-34}	3.59×10^{-33}
PYCARD	6.967	8.183	1.217	0.582	5.48×10^{-28}	1.83×10^{-27}
IL18	7.088	8.250	1.163	0.470	8.47×10^{-19}	2.12×10^{-18}
AIF1	9.033	9.904	0.871	0.382	6.36×10^{-13}	1.27×10^{-12}
IL1B	6.142	7.441	1.299	0.357	1.63×10^{-11}	2.72×10^{-11}
TLR4	9.859	9.571	-0.288	-0.188	4.06×10^{-4}	5.80×10^{-4}
TNF	4.616	4.283	-0.333	-0.073	0.167	0.209
NLRP3	7.090	7.040	-0.051	-0.063	0.236	0.263
CSF1R	11.563	11.663	0.100	0.037	0.488	0.488

*Median difference = GBM median - LGG median. Cliff's δ is also oriented as GBM minus LGG; positive values indicate a tendency toward higher expression in GBM. P-values are two-sided Mann-Whitney U tests; adjusted p-values use the Benjamini-Hochberg procedure across ten genes. Values are rounded for presentation.

4. Discussion

In this exploratory comparison of TCGA lower-grade glioma (LGG) and glioblastoma (GBM) primary tumour samples, several selected neuroinflammation-associated genes showed differences in their dataset-provided expression values between groups. CXCL10 showed the largest observed difference, followed by CASP1 and PYCARD among the genes with higher median values in GBM. IL18, AIF1 and IL1B also had higher medians in GBM, whereas TLR4 had a lower median. TNF, NLRP3 and CSF1R did not meet the adjusted significance threshold.

These findings represent group-level associations and do not establish that any gene causes differences between tumour types. Bulk-tumour expression data combine signals from tumour cells and non-tumour cells in the tissue. Consequently, observed differences may reflect variation in cellular composition as well as differences in gene expression within particular cell types. The present analysis cannot identify the cellular source of each signal.

Several limitations are important. First, LGG and GBM were compared as separate tumour groups; this analysis did not follow individual tumours over time and therefore cannot demonstrate progression from LGG to GBM. Second, the comparisons were not adjusted for clinical or molecular covariates that may differ between groups. Third, the ten genes were selected as candidates for an exploratory analysis, rather than as a comprehensive representation of neuroinflammation. Finally, the expression values were obtained from the UCSC Xena dataset, whose metadata describe the expression unit as $\log_2(\text{norm_count} + 1)$. The analysis uses these dataset-provided values; the present comparison does not independently validate the upstream normalization procedure.

Future work could assess the robustness of these findings by adjusting for relevant clinical and molecular covariates, testing the patterns in an independent dataset, and documenting software and package versions to improve reproducibility. Cell-type-resolved or single-cell data could also help investigate the cellular sources of the observed expression patterns.

5. Conclusion

This exploratory comparison identified differences in the dataset-provided expression values of several selected neuroinflammation-associated genes between TCGA lower-grade glioma (LGG) and glioblastoma (GBM) primary tumour samples. CXCL10 showed the largest observed difference, and six genes had higher median values in GBM, whereas TLR4 had a lower median. These findings describe associations between tumour groups and do not establish causation, identify the cellular origin of expression signals, or demonstrate longitudinal tumour progression. Further analyses using independent datasets, relevant clinical and molecular covariates, and cell-type-resolved data are needed to assess the robustness and biological context of these findings.

Data Availability

The study used publicly available TCGA lower-grade glioma and glioblastoma gene-expression and clinical data accessed through UCSC Xena. The expression dataset was

[TCGA.GBMLGG.sampleMap/HiSeqV2](#), with metadata-reported expression units of $\log_2(\text{norm_count} + 1)$ and dataset version 2017-09-08. The dataset is available at:

<https://xenabrowser.net/datapages/?dataset=TCGA.GBMLGG.sampleMap%2FHiSeqV2&host=https%3A%2F%2Ftcga.xenahubs.net>. Data accessed October 4, 2026.

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